INCH-POUND
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(See 6.9)

MILITARY SPECIFICATION

BURNER, CARBON MONOXIDE AND HYDROGEN, CATALYTIC TYPE (MARK IV E) NAVAL SUBMARINE USE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the design and construction of Mark IV E equipment for removing carbon monoxide and hydrogen from submarine atmospheres.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

HH-P-31 - Packing and Lagging Material, Fibrous Glass Metallic and Plain Cloth and Tape.

PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4460

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MIL-S-901	- Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-E-917	- Electric Power Equipment, Basic Requirements (Naval Shipboard Use).
MIL-G-1149	- Gasket Materials, Synthetic Rubber, 50 and 65 Durometer Hardness.
MIL-E-2036	- Enclosures for Electric and Electronic Equipment.
MIL-C-2212	- Controllers, Electric Motor A.C. or D.C., and Associated Switching Devices.
MIL-M-7793	- Meter, Time Totalizing.
MIL-A-8625	- Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-F-13785	- Filter Material.
MIL-I-16411	- Insulation Felt, Thermal, Glass Fiber.
MIL-M-17060	- Motors, 60-Hertz, Alternating Current, Integral-Horsepower, Shipboard Use.
MIL-E-17555	- Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts): Packaging of.
MIL-L-19140	- Lumber and Plywood, Fire-Retardant Treated.
MIL-C-21665	- Catalyst, Carbon Monoxide and Hydrocarbon Oxydizing (For Use in an Air Purifying Device Aboard Submarines).
MIL-H-22577	- Heating Elements, Electrical: Cartridge, Strip and Tubular Type.
MIL-H-46855	- Human Engineering Requirements for Military Systems, Equipment and Facilities.

STANDARDS

MILITARY

MIL-STD-167-1	- Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-278	- Welding and Casting Standard.
MIL-STD-471	- Maintainability Verification/Demonstration/ Evaluation.
MIL-STD-681	- Identification Coding and Application of Hookup and Lead Wire.
MIL-STD-740-1	- Airborne Sound Measurements and Acceptance Criteria of Shipboard Equipment.
MIL-STD-740-2	- Structureborne Vibratory Acceleration Measurements and Acceptance Criteria of Shipboard Equipment.
MIL-STD-769	- Thermal Insulation Requirements for Machinery and Piping.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

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2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 333 - Standard Specification for Nickel-Molybdenum Alloy (UNS N1026) Plate, Sheet, and Strip.
B 335 - Nickel-Molybdenum Alloy (UNS N1026) Rod, Specification.
B 424 - Nickel Iron Chromium Molybdenum Copper Alloy (UNS N08825 and UNS N08221) Plate, Sheet, and Strip.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications), the text of this specification takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.5) in accordance with 4.3.

3.2 General description. The burner shall pick up ambient air in the submarine, remove carbon monoxide and hydrogen, and return the air to the space. The air shall flow from the space through a fiber filter, the fan, the inlet path of the heat exchanger, the heater, the catalyst bed, the outlet path of the heat exchanger, the aftercooler and then to the space. These components shall be arranged in a compact assembly for quick and convenient renewal of the catalyst, filter, aftercooler, and heaters. An automatic temperature control system shall be incorporated to control and indicate the temperature within the catalyst bed at 600 ± 10 degrees Fahrenheit (°F). The control system shall indicate an excessive temperature of 650 ± 10°F (as occurs when hydrogen is being burned). A red light shall be provided to indicate this over-temperature. The control system shall also automatically shut down the burner when the temperature reaches 675 ± 10°F.

3.3 Hydrogen burning. When there is hydrogen in the incoming air, combustion on the catalyst will cause a temperature rise in the ratio of about 150°F for 1 percent hydrogen. A manual means of bypassing a portion of the incoming air around the heat exchanger shall be provided to keep the temperature from exceeding 675°F when burning hydrogen in any concentration up to 2 percent.

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3.4 Capacity. With the filters and catalyst installed, the equipment shall be constructed for a minimum air flow of 500 standard cubic feet per minute (std ft^3/min) with an external pressure differential of plus 2.5 inches water gauge.

3.5 Physical configuration. The complete unit shall be constructed of subassemblies or components which will pass through a 25-inch diameter hatch. Sections shall also pass through a 20- by 38-inch door with 10-inch radius corners. The overall dimensions of the unit, including mounting, shall fit into a cubic volume of 39 by 39 inches in width and 67 inches in height. The general arrangement shall be satisfactory to the Command or agency concerned. The control panel and instrument panel shall be mounted on the front of the unit. The unit shall be mounted on six 7E450 mounts: front to rear spacing shall be 34 ± 1/32 inches, with front and rear center mount on centerline of unit and 14-3/4 ± 1/32 inches to each adjacent mount. Mounts shall be 18-7/8 ± 1/32 inches from base of unit to the foundation base and shall be level within plus or minus 3 degrees from the mount plane. Baseplate cutouts if required for all 7E450 mounts shall be 4.1875 to 4.3125 inches in diameter. The weight of the completed unit including catalyst shall be less than 2320 pounds. The design or actual loading of the internal resilient mounts shall be less than the nominal rating; thus, the sustained load of 7E450 mounts shall be less than 450 pounds. The temperature of the resilient mounts at the hottest point shall be less than 150°F. This shall be accomplished without cooling water. Access to the unit for maintenance and operation shall be limited to the front and both sides of the plant. The space required for maintenance and operation shall be limited to 24 inches in front and 12 inches on each side of the plant. The heater assembly shall be removable from the front of the unit (see 6.3 and appendix B).

3.6 Noise and vibration.

3.6.1 Structureborne noise. The structureborne noise shall conform with requirements of type 3 of MIL-STD-740-2 (see 4.6.1).

3.6.2 Airborne noise. The airborne noise output of the complete unit shall conform to the requirements of MIL-STD-740-1 except the sound pressure level limits shall be as follows (see 4.6.2):

Octave band (hertz (Hz))

	37.5 75	75 150	150 300	300 600	600 1200	1200 2400	2400 4800	4800 9600
dB	90	85	80	80	75	75	75	75

3.6.3 Vibration. The burner shall not be damaged by internally excited vibrations, or by environmental vibrations specified in MIL-STD-167-1, type I, for frequencies up to and including 33 Hz (see 4.6.3).

3.7 Shock resistance. The completed assembly shall pass the high impact shock tests specified in MIL-S-901 for grade A, class II equipment (see 4.6.4). It shall be constructed for installation on Navy type resilient mounts with a snubber arrangement.

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3.8 Electrical components. Electric components shall conform to the requirements of MIL-E-917 and enclosures shall be dripproof in accordance with MIL-E-2036. Power supply shall be 440 volt, three phase, 60 Hz. Electrical components shall operate continuously over the temperature range specified in MIL-E-917. Wiring shall be in accordance with MIL-STD-681. Panel wiring shall have terminal markers at both ends; wires less than 12 inches shall be marked in the center only. The control panel access door shall be hinged. The control panel cover shall be equipped with a means to lock the cover in the open position. Provision shall be included for grounding of the burner in accordance with MIL-E-917 at a single location.

3.8.1 Motor. The electric motor shall conform to design B of MIL-M-17060 and the following:

(a) Type: Squirrel-cage induction.
(b) Service: A.
(c) Bearings: Ball.
(d) Duty: Continuous.
(e) Insulation: Class B or F with temperature rise for class B in accordance with MIL-M-17060.

(a) Type: Squirrel-cage induction.
(b) Voltage: 440 volts, 60 Hz.
(c) Enclosure: Totally enclosed fan cooled.
(d) Horsepower: Three.
(e) Speed: As required.
(f) Bearings: Ball.
(g) Duty: Continuous.
(h) Service: A.
(i) Ambient: 122°F (50 degrees Celsius (°C)).
(j) Insulation: Class B, F, or H.
(k) Temperature: At rated loads shall not exceed class B rises for 122°F (50°C) ambient.
(l) Motor shall be for submarine service.
(m) Motor balancing ring required.

Motor balancing rings are required except the fan hub may be used as a balancing ring.

3.8.2 Controllers. Motor starting controllers shall conform to MIL-C-2212 and the following:

(a) Duty: Continuous.
(b) Operation: Magnetic.
(c) Protection: Undervoltages - low voltage protection.
(d) Overload: Thermal overload relay.

3.8.3 Heating elements. Heating elements shall be hermetically sealed in accordance with type I of MIL-H-22577 except the heater tolerances shall be within plus or minus 5 percent. Special attention shall be given to terminal construction and materials in order to ensure easy removal of terminal connections after

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continuous operation without damage to the heating element. Terminal connections shall be of a bus bar type arrangement. The watt density shall not exceed 25 watts per square inch. Individual heaters shall be replaceable in the heater assembly.

3.8.4 Heater controls. The heaters shall be automatically controlled by a temperature indicating controller to maintain the temperature at 600 ± 10°F. Controllers shall include a manual switch so that heaters can be turned off when the fan is running. The heater shall be sealed and shall be removable and replaceable through an access plate in the catalyst bed section of the burner. Heaters shall be protected against overloads. "Heaters On" condition shall be indicated by an amber indicator light on the temperature controller.

3.8.5 Control circuit. The burner control circuit shall, in addition to requirements of 3.8.1 through 3.8.4, consist of a 440/110 transformer, relays in accordance with MIL-C-2212, thermoswitches for indication of excessive temperatures at 650 ± 10°F, fuses, an adjustable temperature control device with its own sensor, of the same type as the temperature indicating controller, to automatically shut down the machine at 675 ± 10°F and a panel mounted override switch. An audible alarm shall be installed in the unit to indicate when a shut down has occurred.

3.8.6 Electrical resistance. Resistance to ground shall be not less than 2 megohms.

3.9 Service connections. Unless otherwise specified (see 6.2), the service connections and their location shall be in accordance with table I.

TABLE I. Service connections. 2/, 3/, 4/, 5/

Service connection	Location and orientation 1/	Size	Type
Water inlet	16-9/16 inches left of ℄ of plant 11-3/16 inches to rear of ℄ of plant 36-5/8 inches elevation Opening: facing up	1 inch nps	----
Water outlet	12-3/16 inches left of ℄ of plant 11-3/16 inches to rear of ℄ of plant 5-7/8 inches elevation Opening: facing down	1 inch nps	----

See footnotes at end of table.

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TABLE I. Service connections - Continued.

Service connection	Location and orientation 1/	Size	Type
Air inlet Centerline	14 inches left of ℄ of plant 7-1/4 inches to rear ℄ of plant 67 inches elevation Opening: facing up	6 inches wide by 22 inches long	Rectangular opening
Air outlet Centerline	17-5/8 inches left of ℄ of plant At ℄ of plant 29-5/8 inches elevation Opening: facing left	5 inches wide by 18 inches long	Rectangular opening

1/ Tolerance for location of fittings shall be plus or minus 1/8 inch.
2/ Elevations are from lower edge of base.
3/ Right and left directions are the observer's right and left while looking at the front of the plant.
4/ Openings shall be parallel to principal plants.
5/ Dimensions which position the path of flow are to the centerline of the fitting.

3.10 Components.

3.10.1 Filters. The material for the fiber filter shall conform to type 5 of MIL-F-13785. The filter arrangement shall be such that no bypass occurs. The filters shall be quickly and easily demountable for servicing. The filter access door shall be retained by quick opening fasteners.

3.10.2 Heat exchanger. The heat exchanger shall transfer, to the inlet air, not less than 72 percent of the increased heat load of the air leaving the catalyst section. The heat exchanger shall be such that any of its surfaces which come in contact with the incoming air shall not come in contact with the air leaving the catalyst section. The heat exchanger flange surfaces and the mating flange surfaces shall be machined surfaces with a roughness less than 63 micro-inches root mean square. The heat exchanger shall prevent any leakage between passes in excess of 20 (std ft^3/min) at 15 inches water gauge differential. The heat exchanger core and seals shall be replaceable. An inspection port shall be provided at the outlet of the first pass. The heat exchanger shall be a plate type with a plate thickness of not less than 0.013 inch.

3.10.2.1 After cooling. A water cooling coil shall be provided on the discharge side of the heat exchanger. The capacity of the external cooling coil shall be such that the temperature of air leaving the external cooling coil shall not exceed 120°F when 2 percent hydrogen is being burned in the catalyst. The external cooling coil shall be part of the burner but shall be removable as an individual unit and shall be completely accessible from the exterior of the burner. The cooling coil shall be so arranged that condensation will not be able

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to drain back into the burner. This performance shall be based on cooling with not less than 15 gallons per minute of 45 ± 2°F fresh water. Water passages shall be tested for tightness and strength at 225 pounds per square inch (lb/in^2). The aftercooler shall operate continuously at 500°F inlet temperature. The cooling water inlet and outlet shall terminate in a 1-inch nominal pipe size monel stub end. The air from the aftercooler shall discharge from the top of the cooler through a 5- by 18-inch opening in the aftercooler cover plate supplied with the burner. The aftercooler shall be provided with an access port to permit flushing and cleaning of the lower portion of the heat exchanger second pass outlet. Opening shall have dimensions not less than 4 by 12 inches. Access port cover plate shall be easily removable, provided with a gasket, and fastened in a manner such that the aftercooler casing is not penetrated. The aftercooler shall be a tube and fin type with a water tube wall thickness not less than 0.028 inch.

3.10.2.2 Insulation and gaskets. Hot external surfaces from the inlet of the heat exchanger through to the outlet of the heat exchanger shall be insulated and lagged, in accordance with MIL-STD-769 and MIL-I-16411 to reduce the insulation surface temperature to less than 150°F. A full face gasket, conforming to type I of HH-P-31 shall be used for the catalyst access plates, the heater assembly flange, and between the heat exchanger and catalyst section. Neoprene gaskets shall be in accordance with MIL-G-1149, class 1. The aftercooler gaskets and the adjacent heat exchanger corner gasket shall be acid resistant and shall not absorb water. Antiseize and sealing compounds shall not be used in the catalyst chamber.

3.10.2.3 Drainage. The catalyst chamber shall preclude any accumulation of liquid in the catalyst bed. A drain plug shall be provided at the low point of the equipment.

3.10.3 Catalyst and catalyst bed. The catalyst shall be in accordance with MIL-C-21665 (mesh size 3 to 8 mesh). The catalyst bed shall have a volume of not less than 1 cubic foot. The velocity through the bed shall not exceed 150 feet per minute. The bed shall be constructed so that no bypassing of air can occur upon settling of the catalyst up to 10 percent. The temperature of the air across the face of the catalyst bed shall not vary from 600°F by more than minus 30 plus 50°F (without bypass operation when measured in accordance with 4.6.7). If gaskets are used they shall withstand up to 800°F. The contractor shall supply catalyst necessary for performing the first article tests (see 4.3) and quality conformance inspection (see 4.4). Catalyst for production units will be furnished by the Government at the unit's points of destination. Means shall be provided to add catalyst from either side of the catalyst bed. The catalyst shall be retained by Incoloy 825 in accordance with ASTM B 424 perforated plate that shall withstand the temperatures and temperature cycling encountered without distortion.

3.10.3.1 Air flow. A pressure gauge and necessary pitot tube arrangement shall be provided for the purpose of determining the air flow through the unit. The accuracy of the air flow measuring system shall be within plus or minus 5 percent. The gauge shall also be used for determining the pressure drop through the catalyst bed, the inlet filter, the heat exchanger inlet side and the heat exchanger outlet-aftercooler. An instrument mounting bracket shall be provided at

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the front of the unit containing the temperature indicating controller, the pressure gauge and valve manifolding. Panel cut-outs shall be provided to facilitate maintenance. Test plugs shall be provided to check the pressure gauge calibration. The pressure gauge shall withstand the maximum pressure developed by the blower. Corrosion resistant steel (CRES) tubing shall be used. Valves and fittings may be brass. Each valve shall be operated individually.

3.10.3.2 Temperature indicating controllers. A temperature indicating controller shall be provided to indicate and control the temperature of the air at the leading edge of the catalyst bed. An independent high temperature adjustable control shall be provided to shut down the unit on overtemperature. A manually operated switch for electrical override of the shutdown shall be provided. The switch shall be panel mounted. Both sensors shall be thermocouple type and shall be installed in a common sheath unless otherwise accepted by the Naval Sea Systems Command (NAVSEA). The sensor location shall be fixed. The sensor and controllers shall operate continuously at temperatures up to 1000°F and shall be fail-safe. The range of the temperature indicator shall be 0 to 800°F. The temperature control system shall be repairable without removing any other component. The temperature indicator shall be an analog readout type and have the following features:

Scale length - 5 inches
Rate time - 0.5 inch
Reset time - 5 minutes
Indicator accuracy - plus or minus 1 percent full scale
Ambient temperature operating range - 32 to 160°F
Set point accuracy - plus or minus 0.25 percent full scale

3.10.4 Time totalizing meter. The plant shall contain a time totalizing meter in accordance with MIL-M-7793. It may be incorporated in lower assemblies if sufficient operational and cost effectiveness can be established by NAVSEA.

3.10.5 Support and test equipment. The contractor shall establish the requirement for support and test equipment at the operational, intermediate, and depot level for repair, overhaul, performance monitoring, alignment, calibration, and preventive maintenance.

3.11 Material. The prohibited materials and metal requirements of MIL-E-917 shall apply. Structural parts shall be corrosion-resistant metals as specified in MIL-E-917 or as specified herein. In addition, aluminum protection by anodic treatment in accordance with MIL-A-8625 or equivalent, may be used. The material for standard parts shall be subject to acceptance by NAVSEA. The heat exchanger and the complete aftercooler including the cover shall be Hastleloy C-276 a nickel-molybdenum alloy (UNS N1026) in accordance with ASTM B 333 or ASTM B 335. The lower heat exchanger flange adjacent to the aftercooler, and the lower corner sealing wedges shall be Hastleloy C-276 a nickel-molybdenum-chromium alloy (UNS N1026) in accordance with ASTM B 333 or ASTM B 335 and the lower heat exchanger casing duct wherever subject to the possibility of condensate leakage shall be nickel-iron-chromium-molybdenum-copper alloy in accordance with

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ASTM B 424. The heat exchanger casing and catalyst section from the outlet of first pass of the heat exchanger shall be CRES. The heat exchanger and catalyst mounting bracket shall be CRES. The compound mass may be fabricated of ferrous metals when protected for corrosion-resistance in accordance with MIL-E-917. When any of the material is formed by any method which will reduce its corrosion-resistance the material shall be treated to restore it to its original condition. Welding shall be in accordance with MIL-STD-278. Nuts and bolts which are used on the complete unit shall be stainless steel of the 300 series. Other mechanical and electrical components shall conform to the various applicable specifications as specified herein. If components are used which are not covered by these specifications, they shall conform to the standards of Underwriters Laboratories Incorporated.

3.11.1 Mercury. Materials shall contain no metallic mercury or mercury compounds and shall be free from mercury contamination (see 6.3). During the manufacturing process, tests, or inspections, the material offered shall not have come in direct contact with mercury or any of its compounds nor with any mercury containing devices employing only a single boundary of containment. (A single boundary of containment is one which is not backed by a second seal or barrier to prevent contamination in event of rupture of the primary seal or barrier.)

3.11.2 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.11.3 Interchangeability. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance and strength.

3.12 Workmanship. Welds shall be free of burrs and there shall be no excessive warpage in the final assemblies. Flange faces shall be within plus or minus 1/64 inch co-planarity, clean, smooth, and free of pits, burrs and other defects. Sharp edges shall be rounded. Bolt holes shall be aligned within plus or minus 1/32 inch. Bolt holes shall be not more than 1/32 inch oversized.

3.13 Human engineering. The plant shall incorporate the human engineering requirements of MIL-H-46855 and MIL-STD-1472.

3.14 System safety program. The contractor shall develop and maintain an effective system safety program that is planned and integrated into all phases of design, production, and testing of the equipment (see 6.3). The system safety program shall provide a disciplined approach to identify hazards and prescribe corrective actions in a timely cost effective manner in accordance with the system safety precedence established in MIL-STD-882.

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3.14.1 Safety design review. Safety shall be an integral part of all design reviews held for all the equipment, subsystems, and components. The contractor shall conduct system safety program reviews. Where possible, the system safety program reviews shall be conducted as part of the overall program review to assess the status of compliance with the overall safety objectives. This review shall identify any deficiencies of the system with respect to safety and provide guidance for further analysis or design effort which may be required. Qualified contractor system safety personnel shall attend these design reviews. NAVSEA shall be notified prior to each system safety program review, to permit participation by Navy personnel.

3.14.2 Safety analyses. Safety analyses shall be performed to identify hazardous conditions for the purposes of their elimination or control. Analyses shall be made to examine the equipment, subsystems, components and their interrelationship to include logistic support, training, maintenance, and operational environments.

3.15 Reliability and maintainability.

3.15.1 Reliability. The specified mean-time-between-failures (MTBF) of the plant shall be not less than 1500 hours (see 4.6.8). A failure is defined as when either of the following conditions occur:

(a) An unavoidable, forced shut-down of the plant, that cannot be restored without some degree of disassembly to repair or replace an internal element. Include any requirement for maintenance during test not specifically identified as scheduled (preventive) maintenance prior to initiation of the test.
(b) The plant ceases to function or its performance degrades below the requirements of this specification (see 6.3 and appendix A).

3.15.2 Maintainability. The mean-time-to-repair (MTTR) of the plant shall be not more than 4 hours when determined in accordance with MIL-STD-471 (see 6.3 and appendix A).

3.15.3 Failure mode and effects analysis. The contractor shall perform a failure mode and effects analysis (FMEA) (see 6.3 and appendix A).

3.15.4 Maintenance engineering analysis. The contractor shall perform a maintenance engineering analysis (see 6.3 and appendix A).

3.16 Plates.

3.16.1 Identification and instruction plates. Burner assembly shall be provided with an identification plate not to exceed 3 by 3 inches, an instruction plate and an airflow range plate. The type of marking for the letters shall be Gothic capitals, and numerals and other characters shall be of similar appearance. The size of the letters, numerals and other characters shall be as necessary for easy readability. The plates may be etched, engraved, or cast using either monel, brass, corrosion-resisting steel, or aluminum alloy, or may be of the anodized aluminum photographic process.

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3.16.2 Identification plate. The identification plate shall contain the following information:

(a) Item name.
(b) Model and size (when applicable).
(c) National stock number.
(d) Component identification number (CID).
(e) Contract number.
(f) Manufacturer's name and address.

3.16.3 Instruction plate. The instruction plate shall contain the following information:

(a) Simplified operating instructions.
(b) Range of acceptable air flows and temperatures.
(c) Safety precautions.

3.16.4 Anodized aluminum photographic process. Anodized aluminum plates shall have the photosensitive silver compounds imbedded within the oxide layer. After photographic processing, the image shall be sealed in the oxide layer by chemical treatment of the layer. Stamping of additional marking information shall not be permitted.

3.16.5 Air flow range plate. The air flow range plate shall be tabular (50 ft^3/min increments) or graphic and show the air flow versus the pressure drop for a range of 400 to 600 ft^3/min.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contractor or order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program (see 6.3). The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or suppplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the examination of 4.5 and of the tests specified in 4.6.1 through 4.6.9.

4.3.1 Sample for first article inspection. Prior to beginning production, one sample shall be tested as specified in 4.3.

4.4 Quality conformance inspection. Each burner shall be subjected to the examination specified in 4.5 and the tests specified in 4.6.1, 4.6.2 and 4.6.6 (see 6.3).

4.5 Examination. Each burner shall be subjected to an examination to determine conformance with this specification for design, material and workmanship.

4.6 Tests.

4.6.1 Structureborne noise. Before the shock test (see 4.6.4), the burner shall be subjected to the structureborne noise tests specified in MIL-STD-740-2. After the shock test, the structureborne tests shall be repeated. If the burner does not meet the structureborne noise requirements after the shock test, the defects shall be corrected. It is not required that the burner has to meet the structureborne noise requirement after the burner has been subjected to the shock test before any defects are corrected (see 6.3).

4.6.2 Airborne noise. Before and after the shock test (see 4.6.4), the burner shall be tested for airborne noise as specified in MIL-STD-740-1 (see 6.3).

4.6.3 Vibration. A vibration test shall be performed in accordance with MIL-STD-167-1, type I to determine conformance to the requirements specified in 3.6.3. The testing shall be accomplished with the burner attached without mounts and operating under normal rated conditions. The plant shall demonstrate continuous proper functioning during and subsequent to the specified tests (see 6.3).

4.6.4 Shock. The unit shall be subjected to a high impact shock test in accordance with MIL-S-901 (see 3.7). If the results of the shock test require any major changes in design, the test shall be repeated after the change has been incorporated in the unit (see 6.3).

4.6.5 Factory acceptance. The burner shall be subjected to a series of factory acceptance tests to demonstrate its capacity for eliminating carbon monoxide and hydrogen. No factory acceptance requirement with respect to the removal of carbon monoxide and hydrogen is included in this specification. These tests are for information to the Government (see 6.3).

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4.6.5.1 Factory acceptance and characteristics. After the shock test (see 4.6.4), the burner shall be tested to determine its factory acceptance and characteristics with an 8-hour test for each of the following combinations of gases:

(a) Hydrogen - 0.5 percent, 1.0 percent, 1.5 percent and 1.8 percent.
(b) Hydrogen and carbon monoxide - 1.8 percent H_2 and 100 parts per million (p/m) carbon monoxide (CO).
(c) Carbon monoxide - 100 p/m, 50 p/m.

The tests shall be conducted at a room temperature of 70 $\pm$ 5°F and 45 to 55 percent relative humidity. During each test the thermal, electrical and mechanical operation and the constituency of the input and output air streams shall be monitored. For the information of the Government the air temperature of air leaving the unit shall be measured when the unit is buring 2 percent H_2 and the inlet cooling water temperature is 45 $\pm$ 2°F.

4.6.6 Uniformity. The uniformity of the temperature across the face of the catalyst bed (see 3.10.3) shall be demonstrated. Not less than 24 thermocouples located on approximately 4.5 inch centers in a single vertical plane shall be used. The heat exchanger shall be pressure tested to verify the requirements of 3.10.2. The completely assembled unit less the fan or motor assembly shall be pneumatically pressure tested to 28.0 to 29.0 inches water column. The pressure loss shall not exceed 20 inches water column at the end of 30 minutes (final pressure shall be not less than 8.0 inches water column). The aftercooler water side shall be hydrostatically tested at 180 lb/in^2 and shall show no leaks. The resilient mounts temperature shall be measured to determine the maximum mount temperatures. The center of gravity and mount deflection shall be measured.

4.6.7 Safety testing. Tests shall be approved by the contracting activity.

4.6.8 Reliability demonstration test. The contractor shall subject the equipment to a 720-hour endurance test with no failure (see 3.15.1) (if similar tests have not already been accomplished on a previously supplied identical unit). The following test provision shall apply:

(a) Non-failure events: Failure shall not be considered to have occurred if the event is the result of a cause such as variation of input voltage beyond specified limits, input power interruption, operator error, ambient temperature over specified limit, or other extraneous causes, determination of which shall be subject to acceptance by the contracting activity (see 6.3).

The tests, or portion thereof specified in 4.6.5 may be run concurrently with the endurance test; and shall be conducted using atmospheric air at ambient conditions. If the tests of 4.6.5 are conducted concurrently with the endurance test those tests shall be met throughout the test period.

4.6.9 Maintainability demonstration test. The contractor shall conduct a maintainability demonstration test to verify the requirements specified in 3.15.2 (see 6.3). Tests shall be approved by the contracting activity.

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4.7 Rejection. Breakage, deformation, or failure to meet the requirements of this specification when examined and tested in accordance with 4.3 shall be cause for rejection.

4.7.1 Failure of reliability and maintainability tests. In the event of inability to reach an "accept" decision, the contractor shall perform an analysis to determine the cause, perform corrective action, and propose appropriate modifications to the equipment. Verification of success of any modifications made shall be accomplished by a period of retest to be determined by the contracting activity.

4.7.2 Mercury contamination. Mercury contamination or reasonable cause to suspect material of being contaminated by mercury shall be cause for rejection.

4.8 Inspection of packaging. Sample packs, and the inspection of the preservation, packing, and marking for shipment, stowage, and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.7).

5.1 General.

(a) Navy shipboard stowage fire-retardant requirements.

(1) Treated lumber and plywood. Unless otherwise specified (see 6.2), all lumber and plywood including laminated veneer material used in shipping containers and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Levels A and B - Type II - Weather resistant.
Category 1 - General use.
Level C - Type I - Non-weather resistant.
Category 1 - General use.

(2) Fiberboard. Unless otherwise specified (see 6.2), fiberboard used in the construction of class-domestic, non-weather resistant fiberboard, cleated fiberboard boxes including interior packaging forms shallmeet the flame spread index and the specific optic density requirements of PPP-F-320 and amendments thereto.

5.2 Preservation, packing, and marking. Carbon monoxide and hydrogen burners and accesories, shall be preserved level A, C, or commercial, packed level A, B, C, or commercial and marked in accordance with MIL-E-17555, as specified (see 6.2) and for levels A, B and C as follows (see 6.3):

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(a) Carbon monoxide and hydrogen burners shall be preserved by level A, method II.
(b) Openings into the burner shall be capped, sealed, or plugged to assure effective action of the desiccant.
(c) Desiccant shall be placed in any interior spaces of the burner. The desiccant shall be distributed within the barrier and shall be secured by taping or tying or otherwise secured to prevent movement, rupture of the desiccant bags, or damage to the barrier.
(d) The method of mounting the burner within the container shall ensure the integrity of the requirements of MIL-STD-740-1 and MIL-STD-740-2.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The carbon monoxide and hydrogen burners described herein are used to remove carbon monoxide and hydrogen from submarine atmospheres.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(c) When first article required (see 3.1).
(d) Location of service connections, if other than as specified (see 3.9).
(e) When fire-retardant treated materials are not required (see 5.1).
(f) Preservation, packing and marking requirements (see 5.2).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.5	DI-V-7002	Provisioning parts list	---
3.5	DI-V-23828	Data, screening: provisioning and other preprocurement	---
3.5	DI-V-7196	Statement of prior submission	---

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Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.5	DI-E-20477	Imaged aperture/ tabulating card	---
3.5 and appendix B	DI-DRPR-80651	Engineering drawings	Level 3
3.11.1	DI-E-2121	Certificate of compliance	---
3.14	DI-SAFT-80102	Safety assessment report	---
3.15.1, 3.15.2, 3.15.3, 3.15.4, and appendix A	DI-MISC-80652	Technical information reports	---
4.1.1	DI-R-4803	Inspection system program plan	---
4.4	UDI-T-23732	Procedures, test	---
4.4	DI-T-5329	Inspection and test reports	---
4.6.1	DI-HFAC-80274	Equipment structureborne vibration acceleration measurements test report	---
4.6.2	DI-HFAC-80272	Equipment airborne sound measurements test report	---
4.6.3	UDI-T-23762	Report, vibration testing	---
4.6.4	UDI-T-23753	Reports, equipment shock test	---
4.6.5		Factory acceptance test procedures	---
4.6.8	UDI-T-23724	Report, failure/ malfunction	---
4.6.8	DI-RELI-80250	Reliability test plan	---
4.6.9	DI-R-2129	Plan, maintainability demonstration	---
4.6.9	DI-R-7113	Report, maintainability demonstration	---
5.2	DI-PACK-80120	Preservation and packing data	---
5.2	DI-PACK-80121	Special packaging instructions (SPI)	---

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.4.1 Technical content. Technical content should include the requirements of appendix C.

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6.5 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the items should be a first article sample (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.6 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.6.1 Spare or repair parts. When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.7 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are required by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.8 Subject term (key word) listing.

Ambient air
Automatic temperature control
Catalyst
Fiber filter
Heat exchanger

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 4460-N043)

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APPENDIX A

TECHNICAL INFORMATION REPORT TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers the technical content requirements that should be included on reports when required by the contract or order. This appendix is mandatory only when, data item description DI-MISC-80652 is cited on the DD Form 1423.

20. APPLICABLE DOCUMENTS

20.1 Government documents.

20.1.1 Standard. The following standard forms a part of this document to the extent specified herein. Unless otherwise specified, the issue of this document is that listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

STANDARD

MILITARY

MIL-STD-470 - Maintainability Program Requirements (for Systems and Equipments).

(Application for copies should be addressed to the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

30. REPORTS

30.1 Failure mode and effects analysis. When required by the contract or order, the failure mode and effects analysis shall contain the following information: The analysis shall be conducted to the functional subassembly level except for those failure modes with a level of severity of 4. For these failure modes, the analysis shall be extended to the individually replaceable item level. The FMEA shall identify the design provisions that have been incorporated to inhibit or limit the frequency of occurrence of failure modes with a level of severity of 3 or 4.

30.2 Maintenance engineering analysis. When required by the contract or order, the maintenance engineering analysis shall contain a detailed list of equipment assemblies, sub-assemblies, modules and special support equipment to be analyzed.

30.3 Reliability program plan. When required by the contract or order, the reliability program plan shall contain the following information:

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(a) Reliability management procedures:

(1) Reliability organization.
(2) Management and control.
(3) Subcontractor and contractor reliability program.
(4) Program review.
(5) Status reports.

(b) Reliability design procedures:

(1) Design techniques.
(2) Reliability analysis.
(3) Parts reliability.
(4) Failure mode and effects analysis.
(5) Reliability critical items.
(6) Effects of storage, shelf-life, transportation, handling and maintenance.
(7) Design reviews.

30.4 Maintainability program plan. When required by the contract or order, the maintainability program plan shall contain the following elements of the detailed requirements of MIL-STD-470:

(a) Perform maintainability analysis.
(b) Prepare inputs to the detailed maintenance plan.
(c) Perform design trade-offs.
(d) Predict maintainability values.
(e) Incorporate and enforce maintainability requirements.
(f) Integrate other items.
(g) Participate in design reviews.
(h) Establish data collection, analysis, and corrective action system.
(i) Perform maintainability demonstration.
(j) Prepare maintainability status reports.

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APPENDIX B

ENGINEERING DRAWINGS TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers the technical content requirements that should be included on drawings when required by the contract or order. This appendix is mandatory only when data item description DI-DRPR-80651 cited on the DD Form 1423.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. DRAWINGS

30.1 Drawings. When required by the contract or order, drawings shall contain the following information:

(a) Evaluation of performance and quality of the equipment against requirements of this specification.
(b) Evaluation of operational and personnel safety.
(c) Acquisition of parts by the supply system.
(d) Installation, operation, maintenance and repair of the equipment and all components thereof without contractor's assistance.
(e) Information to appear on the instruction plate and the arrangement of the information.

30.2 Dimensional outline drawings. The outline drawings of the assembled equipment shall be complete so that they can be used for installation guidance. Drawings shall show the following:

(a) Attached auxiliaries and foundation and mounting dimensional requirements, including access arrangements, panel hinging and pull spaces.
(b) Table of weight of individual components and weight of complete unit.
(c) List of shipbuilder's connections showing size, type and dimensions including methods and sizes of fastenings, dimensions and clearances as required for installation and servicing plus supplementary data as necessary to permit installation without the contractor's assistance.
(d) Radii of gyration of complete net unit about each of the three principal axes.
(e) Vertical center of gravity for each assembly above the lowest extremity of the equipment support.

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30.3 Diagram drawings. Diagram drawings shall include a tabulation diagram, electrical schematic and electrical connection diagrams with certification data. Each diagram shall show by symbolic representation all piping, fittings, valves, electrical wiring and connections required by others for installation. The diagrams shall show the size, material and type of piping, fittings, valves and wiring as applicable. Valves, components, accessories, controls and associated instruments shall be given a piece number and be identifiable with the certification data. Piece numbers shall be shown in a consecutive order around each diagram where practical.

30.4 On board repair parts list. A list of on board repair parts shall be furnished with drawings. This list shall be assigned a manufacturer's drawing number.

30.5 Assembly drawings. Assembly drawings shall include a sectional assembly with a complete list of material, with references to applicable subassembly and detail drawings. The list of material shall include every part required in the assembly, including those parts not required to be detailed. This may necessitate some side or partial views in order to show parts not otherwise shown in the main section view. The list of material shall include an indication of each part to be furnished as an on board repair part (this is not a repair parts list). Assembly drawings shall also include performance data or curves as applicable.

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APPENDIX C

MANUAL TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers the technical content requirements that should be included in manuals when required by the contract or order. This appendix is not a mandatory part of the specification. The information contained herein is intended for guidance only.

20. APPLICABLE DOCUMENTS

20.1 Other Government publication. The following other Government publication forms a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

NAVSEA

PUBLICATION
0938-LP-055-1010 - Burner, CO and Hz, MK IVE, Size 500 CFM, Type I.

(Application for copies should be addressed to the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

30. MANUALS

30.1 Manuals. When required by the contract or order, NAVSEA 0938-LP-055-1010 shall be the basic technical manual. Change pages will be required to update the manual. Configuration control shall be maintained. By using serial numbers and matrices, changes may be documented without losing the original configuration. At a minimum the manual shall include on board repair procedures for component replacement for the following subassemblies: blower/motor, heat exchanger, temperature control system and heater assembly. The manual shall also include internal cleaning procedures for the aftercooler and heat exchanger, calibration, grounding, pressure test procedures and an insulation repair procedure.

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